

Home Credit Default Risk – Machine Learning Project

Problem Statement

- Financial institutions often reject applicants with limited credit history, even if reliable. This project aims to predict the probability of default to enable fair, data-driven lending decisions.

Approach

- Integrated multiple Home Credit datasets: applicant info, previous loans, credit card balances, and payment histories.
- Preprocessing included missing value handling, categorical encoding, scaling, and class balancing.
- Evaluated six supervised models using accuracy, precision, recall, F1-score, and ROC-AUC.

Implementation Overview

- Aggregated features by applicant ID, computing mean, max, and min for numeric fields.
- **Models evaluated:** Logistic Regression, Decision Tree, Random Forest, KNN, SVM, MLP.
- **Best model:** Logistic Regression (Accuracy: 69.97%, ROC-AUC: 0.7629).
- Key features: EXT_SOURCE_2, EXT_SOURCE_3, DAYS_BIRTH.
- **Unsupervised Learning – K-Means + Random Forest**
- K-Means segmented applicants into **four clusters** capturing distinct borrower profiles.
- Random Forest trained per cluster improved overall performance.
- **Cluster-wise Performance:**
- Each cluster shows unique repayment patterns.
- Separate models per cluster capture group-specific patterns, boosting predictive accuracy.

Challenges & Conclusions

- Handling large datasets, optimizing memory, and class imbalance were major challenges.
- Logistic Regression is stable and interpretable for the overall dataset.
- K-Means + Random Forest captures heterogeneous groups, improving predictions.
- Key predictors: external credit scores (EXT_SOURCE_2, EXT_SOURCE_3) and age.
- Combining supervised and unsupervised learning enhances credit risk assessment and financial inclusion.